

API Security Risk Analysis Report

Prepared By:

DEEKSHITHA GEDDADA

Internship:

Future Interns Cyber Security Internship 2026

Date:

27 May 2026

1. Introduction

This report presents a basic API Security Risk Analysis conducted on the JSONPlaceholder public demo API. The assessment was performed using safe and read-only testing methods through Postman. The objective of this analysis was to identify common API security risks such as lack of authentication, excessive data exposure, and weak access control observations.

No exploitation, bypass attempts, or harmful activities were performed during this assessment.

2. Objective

The objective of this assessment was to:

- Analyze a public API safely
- Understand how APIs expose data
- Identify possible API security risks
- Observe authentication and authorization behavior
- Document findings in a professional format
- Suggest remediation recommendations

3. API Tested

API Name

JSONPlaceholder API

API URL

<https://jsonplaceholder.typicode.com>

Endpoints Tested

Endpoint 1

GET /posts

Full URL:

<https://jsonplaceholder.typicode.com/posts>

Endpoint 2

GET /users

Full URL:

<https://jsonplaceholder.typicode.com/users>

4. Tools Used

The following tools were used during the assessment:

- Postman
- Web Browser
- GitHub
- MS Word

5. Scope of Testing

This assessment was limited to:

- Public demo API testing
- Read-only GET requests
- Response analysis
- Header observation
- Documentation review

The following activities were NOT performed:

- Exploitation attempts
- Authentication bypass
- Denial-of-Service testing
- SQL Injection testing
- Brute-force attacks
- Attacks against private systems

6. Methodology

The following methodology was followed during the assessment:

1. Reviewed the public API documentation
2. Sent GET requests using Postman
3. Inspected API responses
4. Observed exposed data fields
5. Reviewed response behavior
6. Identified potential security risks
7. Classified risk severity
8. Prepared remediation recommendations

7. Testing Process

Test 1 — Public Posts Endpoint

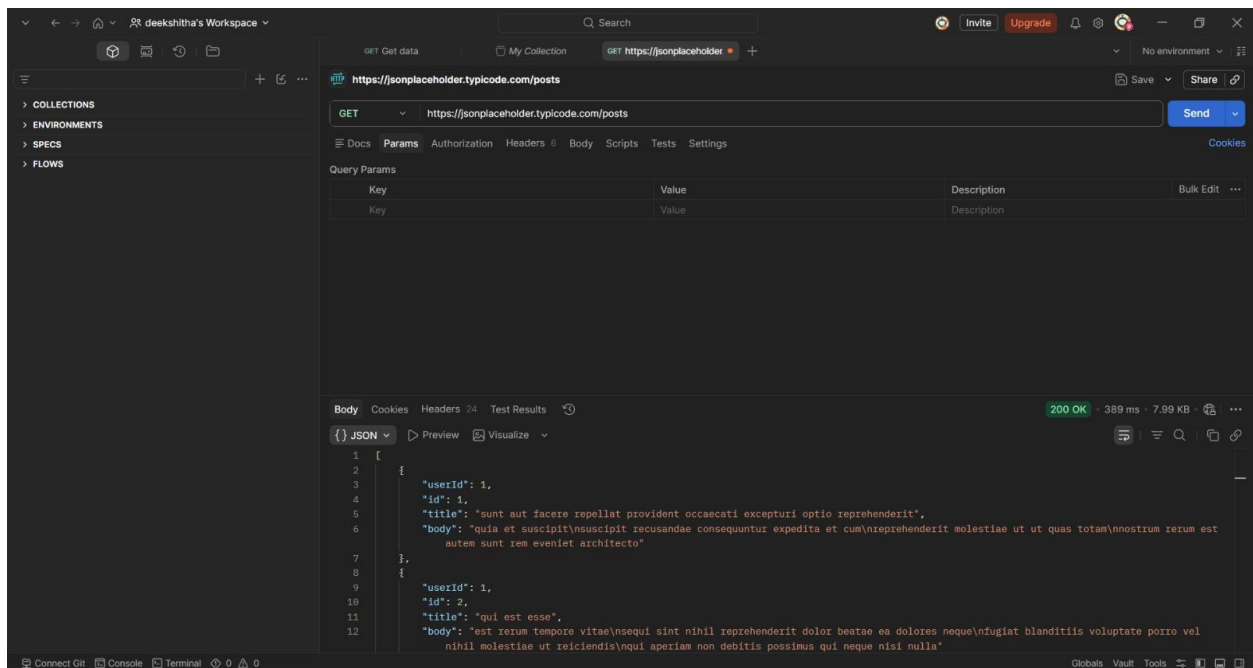
Request

GET <https://jsonplaceholder.typicode.com/posts>

Observation

The API successfully returned post data without requiring any authentication token or user login.

Result



The endpoint was publicly accessible.

Test 2 — Public Users Endpoint

Request

GET <https://jsonplaceholder.typicode.com/users>

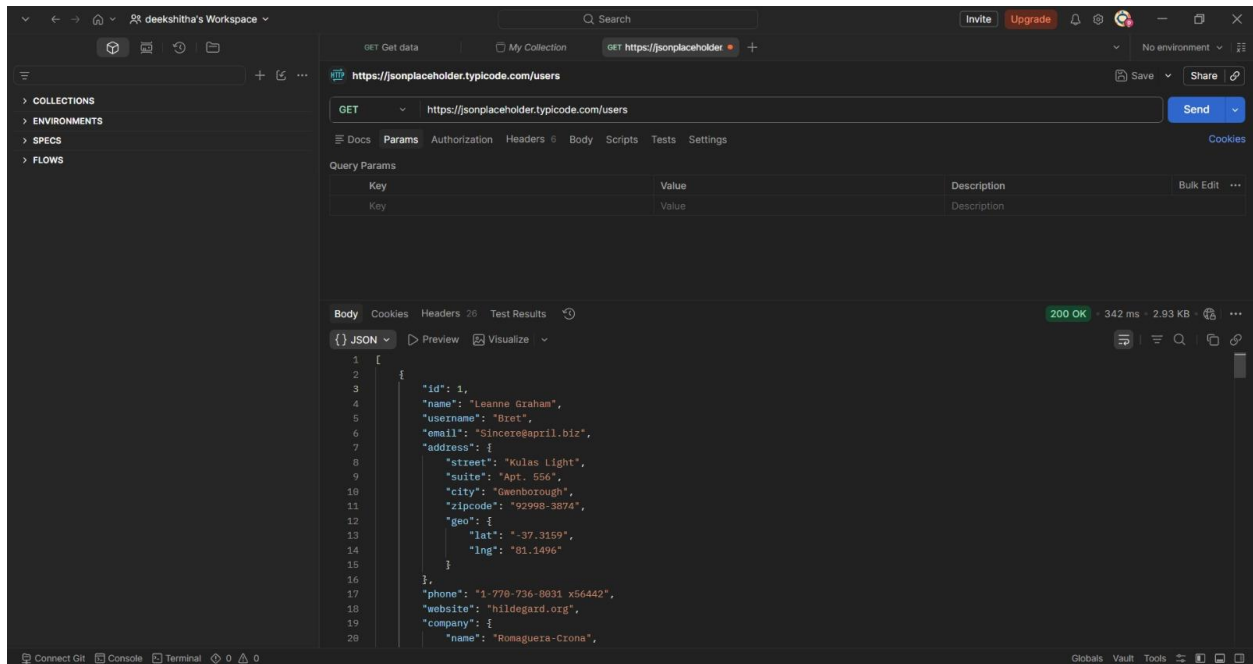
Observation

The API returned user-related information including:

- Names
- Email addresses
- Phone numbers
- Website details

The information was accessible publicly without authentication.

Result



User-related information was exposed publicly through the API response.

8. Findings & Risk Analysis

Finding 1 — Missing Authentication

Risk Severity

Medium

Description

The API endpoints were accessible without authentication or authorization tokens.

Observation

Users could retrieve API data directly through public GET requests.

Business Impact

Publicly accessible APIs may allow unauthorized users or automated tools to collect information from the service.

Recommendation

Implement authentication mechanisms such as:

- API keys
- OAuth tokens
- JWT authentication

Sensitive endpoints should require authenticated access.

Finding 2 — Excessive Data Exposure

Risk Severity

High

Description

The API exposed user-related information publicly in the response.

Exposed Information

- Name
- Email
- Phone number
- Website information

Business Impact

Excessive exposure of user information may create:

- Privacy concerns
- Data collection risks
- User profiling risks
- Reputation impact for organizations

Recommendation

Limit API responses to only required fields and avoid exposing unnecessary user information publicly.

Finding 3 — Weak Authorization Observation

Risk Severity

Medium

Description

Different user data records could be accessed directly through public endpoints.

Observation

User information appeared accessible without ownership verification.

Business Impact

Weak authorization controls in real-world systems could allow unauthorized access to user data.

Recommendation

Implement proper authorization and access control validation for user-specific resources.

9. Overall Risk Summary

Risk	Severity
Missing Authentication	Medium
Excessive Data Exposure	High
Weak Authorization	Medium
Observation	

10. Conclusion

The API Security Risk Analysis identified several common API security weaknesses including lack of authentication, excessive data exposure, and authorization-related observations.

Although the tested API is a public demo environment intended for learning purposes, the identified findings reflect risks commonly observed in real-world applications.

Organizations should implement:

- Proper authentication
- Strong access control
- Minimal data exposure
- Secure API design practices

to reduce security risks and protect user data.

11. Ethical Statement

This assessment was performed ethically and legally using public demo APIs only.

No exploitation, harmful activity, or unauthorized access attempts were conducted during testing.